

Block box approaches to activity sequence modelling

Fred Shone¹ and Tim Hillel¹

¹Department of Civil, Environmental and Geomatic Engineering, University College
London, UK

August 17, 2026

Contact: fredjshone@gmail.com
Presenting in person.

1 Introduction

Typical MATSim runs require an initial activity plan for every agent: a sequence of activities with types and locations, ideally with sensible start times and durations. These input plans are generally required to be representative of current or possible future behaviours by agents. There are two broad approaches to modelling these input plans: (i) sampling-based, and (ii) modelled, typically with an activity-based demand model.

Sampling-based approaches rely on finite available data and so are prone to *lumpiness* and *zero-samples* (i.e. they lack realistic diversity). Sampling also assumes behaviours unchanged from when the data is collected. Modelling approaches can model novel scenarios and more realistically represent diversity, but are slow and expensive to develop or adopt for new scenarios.

We present ActVAE, a deep conditional generative model that replaces or mimics alternative activity-based approaches to activity scheduling (modelling activity participations and their timing). Refer to Shone and Hillel [2026] for full methodological detail and results. Note that current work does not include activity location modelling or mode choice, but these are likely future extensions.

2 Methodology

ActVAE (Figure 1) is a sequence-to-sequence Variational Auto-Encoder with a Conditional Prior (CP-VAE), available in the open-source Caveat¹ project. Schedules are encoded as sequences of activity types with continuous durations and modelled end-to-end using recurrent neural networks, conditioned on individual and household labels such as age, income, employment, public transport accessibility, and so on, as available.

ActVAE is trained on the 2023 UK National Travel Survey (59,265 schedules, 8 activity types, 8 conditioning labels) and evaluated against (i) a bespoke baseline based on CEMDAP, and (ii) an LLM (specifically Gemma3-1B). Results are shown in Table 1.

¹<https://github.com/big-ucl/caveat>

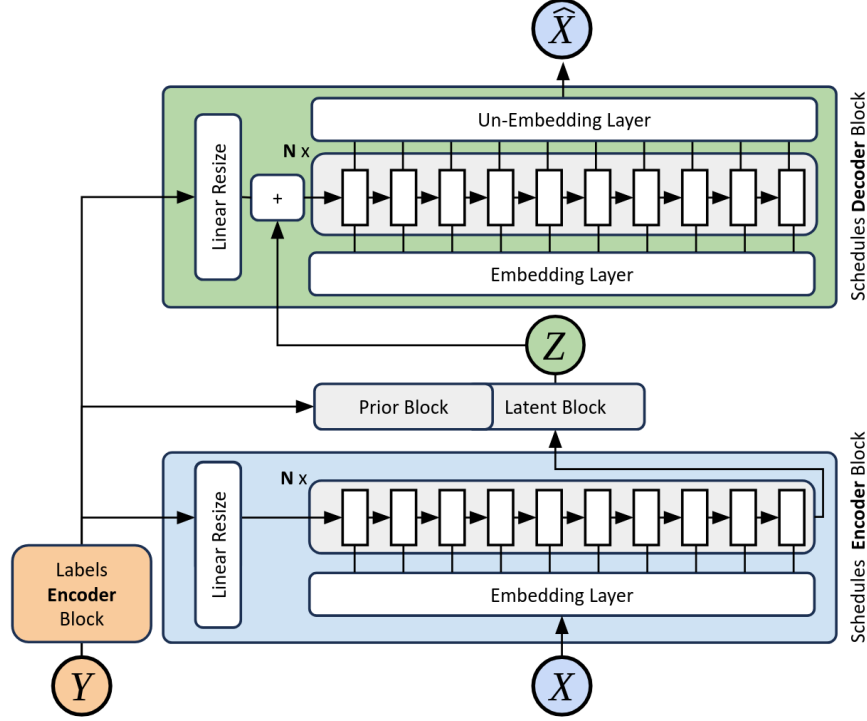


Figure 1: ActVAE architecture: Schedules (X) are passed through an encoder-decoder architecture (both composed of 4x256 LSTM layers). Output schedules ($\hat{X}$) are conditional on labels (Y) via a label encoder (embeddings combined with addition, followed by a single fully connected layer, all of size 32). The encoder-decoder is bottlenecked through a learnt conditional prior over the latent representation (Z). Each LSTM block outputs both a categorical prediction of activity type and a duration. Special start and end-of-sequence tokens are used to define and predict activity sequence length. The prior and latent blocks use fully connected layers to resize inputs and outputs for a latent vector of size 6.

3 Key findings

3.1 Speed/cost

ActVAE trains in around three minutes on a single GPU and generates roughly 100,000 schedules per second, fast enough to produce schedules for a 10% national-scale scenario in around 1 minute. New data, scenarios, or recalibration require no manual re-specification, enabling rapid generation of activity plans for novel scenarios. This speed would also make within-simulation plan modifications viable.

3.2 Realism/distribution

ActVAE performs comparably to baseline approaches at predicting activity participations and their order, but performs better at predicting the more complex distributions of activity timings, including bimodal patterns of escort trips (Figure 2). Note that this realism includes the capturing of, often subtle, conditional distributions of schedules with age, income, accessibility, and so on, detailed in Figures 3 and 4.

Table 1: Evaluation summary

	Baseline*	ActVAE*	LLM	Unit
Speed ↑				
Generation speed	1000	100,000	1	schedules/second
Realism (density estimation distances) ↓				
Participations	0.084	0.083	1.104	rate EMD
Transitions	0.004	0.004	0.008	rate EMD
Timing	0.080	0.047	0.215	days EMD
Probability a schedule is not unique	0.088	0.033	0.423	rate EMD
Probability a schedule is not novel	0.032	0.031	0.042	rate EMD
Feasibility ↓				
Probability not home-based	0.000 †	0.000	0.000	probability
Probability of consecutive**	0.000 †	0.014	0.000	probability

* Means from 5 model runs, variance is <0.0005 throughout.

**Probabilities of adjacent home, work or education activities.

† Zero probability structurally guaranteed by model.

3.3 Feasibility/controllability

ActVAE guarantees temporal consistency but cannot guarantee other types of structural constraints, such as home-based schedules only. However, we show that such constraints can be modelled (from the data alone) with high reliability (>99% of the time), so that they can be reasonably rejected or fixed after generation.

3.4 A note on traditional approaches

We use a baseline scheduling model based on Daily Activity Plan (DAP) prediction, based on CEMDAP [Sener et al., 2006]. The code for our baseline model is available from Composed². DAP, and tour-based approaches more generally, are good at capturing common activity participation patterns and can guarantee structural feasibility. However, this comes at the cost of (i) loss of realistic diversity, particularly of timing and for less common or unusual behaviours, and (ii) speed, as complex model structures require significant time to specify and calibrate.

3.5 A note on behaviour modelling with LLMs

LLMs have some utility as an accessible approach to modelling human behaviours, for example, allowing feasible schedules to be defined and generated using natural language alone. However, we find them to be extremely slow, costly and unrealistic in aggregate. This finding is consistent across many different models, of various sizes and using various approaches, including few-shot and fine-tuned. The code used to explore LLMs as activity scheduling models is available from ActLLM³.

4 Conclusions

We intend ActVAE as a drop-in scheduling component within an activity-based demand modelling framework producing plans for MATSim, particularly where speed of scenario iteration is valuable

²<https://github.com/big-ucl/composed>

³<https://github.com/fredshone/actllm>

or where representing the diversity of real schedules matters. However, ActVAE does not conform to any existing behavioural theory, has no intrinsic interpretability, and cannot be guaranteed to produce feasible schedules. Future work will likely look to extend the approach to full activity plans, with locations and trip information.

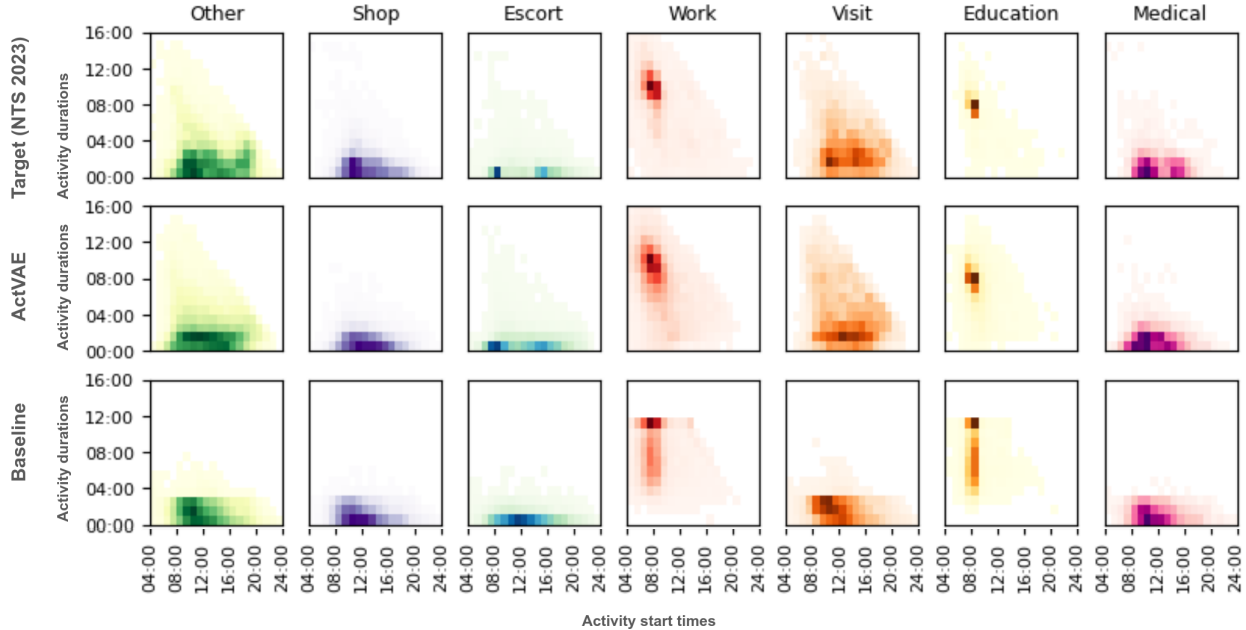


Figure 2: Joint distribution of activity start times and durations, by activity type. ActVAE reproduces the target (NTS 2023) distributions well, including bimodal escort timing. The baseline approach shows some unrealistic artifacts in distribution

References

- Ipek N Sener, Chandra R Bhat, Rachel Copperman, Sivaramakrishnan Srinivasan, Jessica Y Guo, Abdul Pinjari, and Naveen Eluru. Activity-based travel-demand analysis for metropolitan areas in texas: Cemdap models, framework, software architecture and application results. Technical report, Midwest Regional University Transportation Center, 2006.
- Fred Shone and Tim Hillel. Black box behavioural modelling: Predicting human activity schedules with a deep conditional generative approach, 2026. URL <https://arxiv.org/abs/2512.04223>.

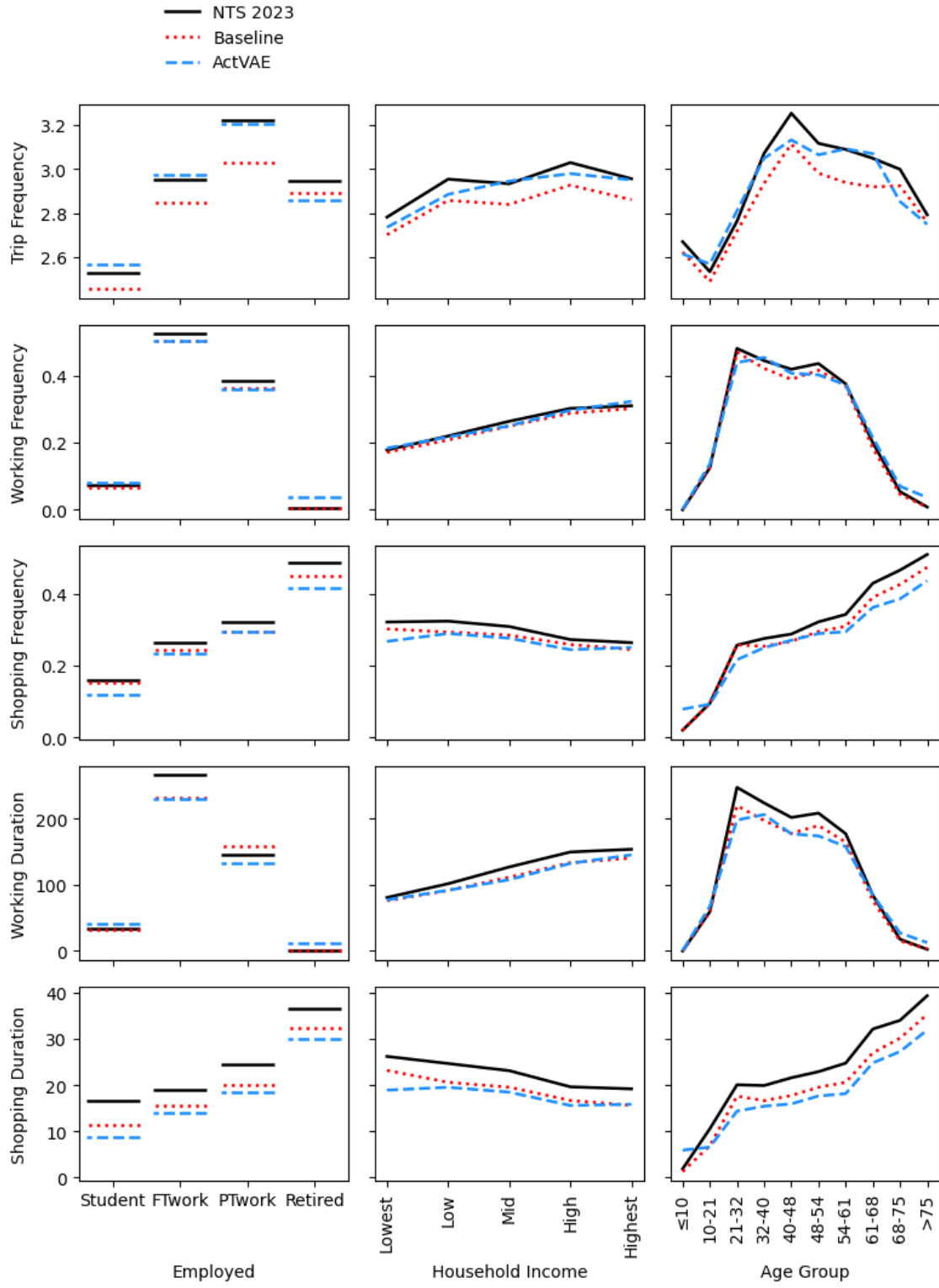


Figure 3: Expected sensitivities of trip frequency, work and shop participations and durations to employment status, household income and age.

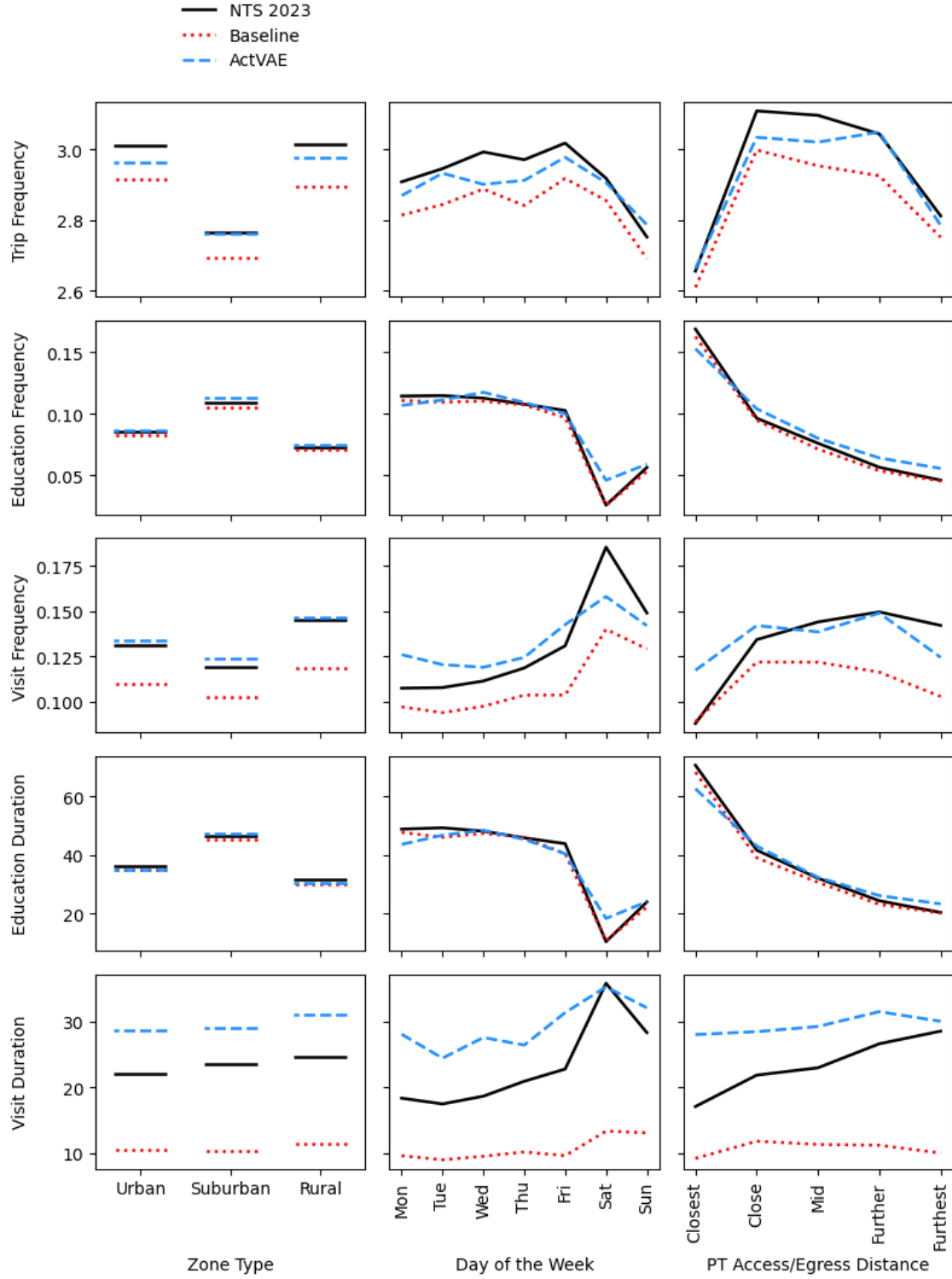


Figure 4: Expected sensitivities of trip frequency, education, and visit participations and durations to household zone type, day of week and PT accessibility (based on distance to nearest stop).